

Exp -13 Artificial Neural Network (ANN)

Program:

```
from sklearn.neural_network import MLPClassifier

X = [
    [25,120],
    [35,140],
    [45,150],
    [30,130],
    [50,160]
]

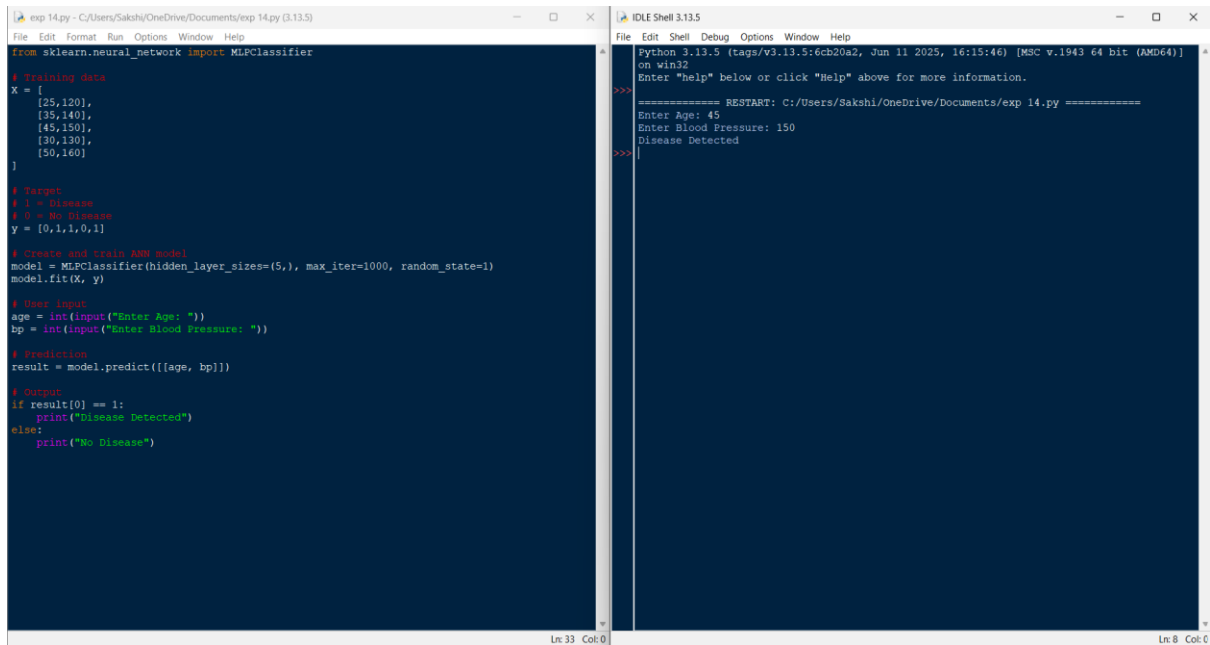
y = [0,1,1,0,1]

model = MLPClassifier(hidden_layer_sizes=(5,), max_iter=1000, random_state=1)
model.fit(X, y)

age = int(input("Enter Age: "))
bp = int(input("Enter Blood Pressure: "))
result = model.predict([[age, bp]])

if result[0] == 1:
    print("Disease Detected")
else:
    print("No Disease")
```

Output:



The image shows a Python IDE with two windows. The left window, titled 'exp 14.py', contains a script for training and testing a neural network classifier. The script imports `MLPClassifier` from `sklearn.neural_network`, defines training data `X` and target labels `y`, trains a model, and then takes user input for age and blood pressure to make a prediction. The right window, titled 'IDLE Shell 3.13.5', shows the execution output, including a restart message and the results of the prediction for age 45 and blood pressure 150.

```
exp 14.py - C:/Users/Sakshi/OneDrive/Documents/exp 14.py (3.13.5)
File Edit Format Run Options Window Help
from sklearn.neural_network import MLPClassifier
# Training data
X = [
    [25,120],
    [35,140],
    [45,150],
    [30,130],
    [50,160]
]
# Target
# 1 = Disease
# 0 = No Disease
y = [0,1,1,0,1]
# Create and train ANN model
model = MLPClassifier(hidden_layer_sizes=(5,), max_iter=1000, random_state=1)
model.fit(X, y)
# User input
age = int(input("Enter Age: "))
bp = int(input("Enter Blood Pressure: "))
# Prediction
result = model.predict([[age, bp]])
# Output
if result[0] == 1:
    print("Disease Detected")
else:
    print("No Disease")
Ln: 33 Col: 0
```

```
IDLE Shell 3.13.5
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Sakshi/OneDrive/Documents/exp 14.py =====
>>>
Enter Age: 45
Enter Blood Pressure: 150
Disease Detected
>>>
Ln: 8 Col: 0
```